Cloud LearnX

Mathematics

Transformations — Practice Worksheet

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Topic: Transformations

Instructions:

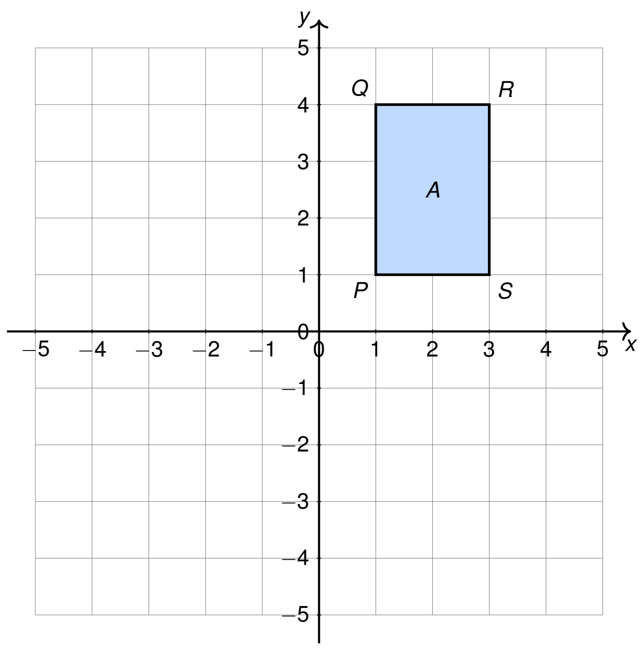
- Answer ALL questions.
- Show ALL working — method marks are awarded for correct steps.
- Write answers in the spaces provided.

Answer ALL ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1 A space agency designer is plotting the flight path markers for a satellite launch simulation on a coordinate grid. Shape *A* represents the position of a rocket module, as shown below.

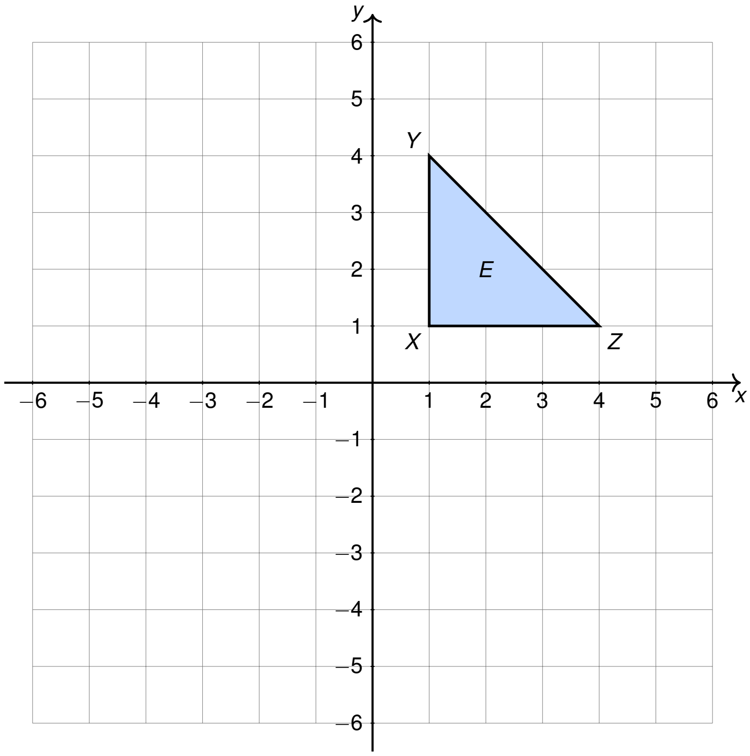


On a copy of the grid, reflect shape *A* in the *x*-axis. Label the image *B*.

(2)

(Total for Question 1 is 2 marks)

2 A space mission planner is designing a symmetrical logo for a satellite launch. The design team draws triangle E on a coordinate grid to represent one half of the emblem, as shown below.



Reflect triangle E in the y-axis.

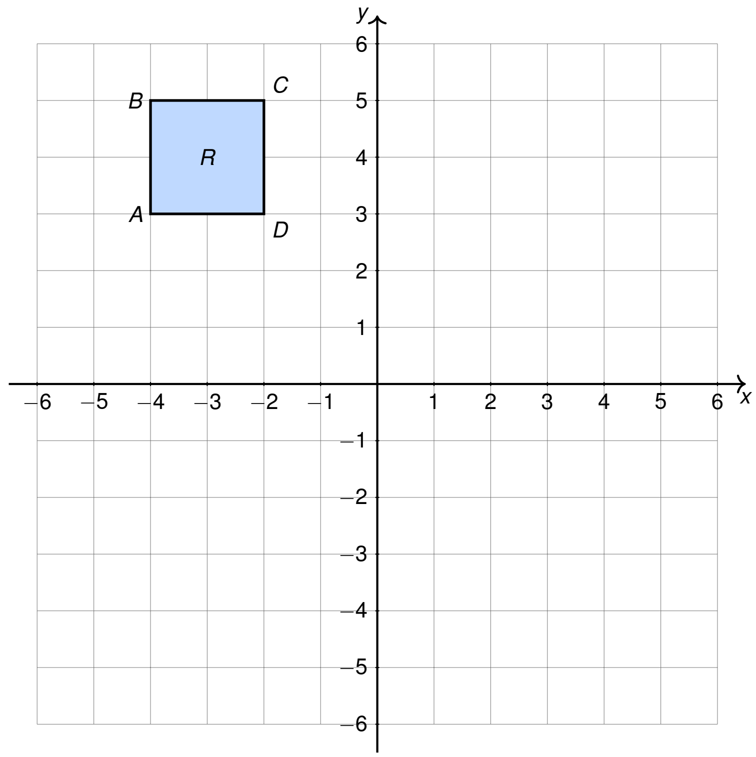
Draw and label the image triangle E' on the grid, showing the coordinates of each vertex clearly.

....

(2)

(Total for Question 2 is 2 marks)

3 A space exploration museum displays a scale model of a rover on a grid map of an exhibition floor. Shape R shows the rover's starting position. The rover is moved along the floor by the translation $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$ to its new display position, shape R'.



On the grid, draw the image of shape R after a translation by the vector $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$.

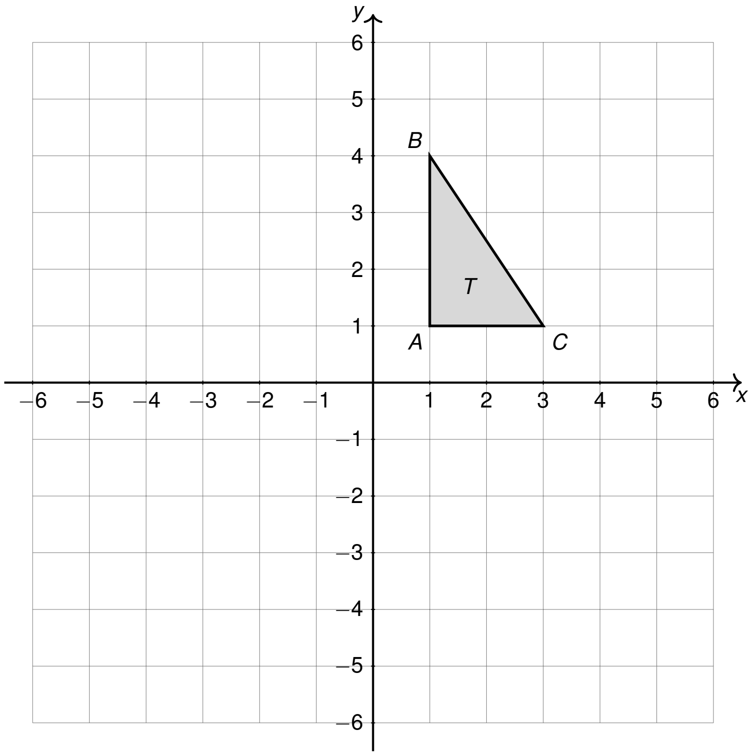
Label the image R'.

....

(2)

(Total for Question 3 is 2 marks)

4 A space agency uses a computer model to track the orientation of a triangular solar panel, T, on a satellite. On the grid below, the panel is rotated 90° clockwise about the origin O to a new orientation, T'.



On a copy of the grid, rotate triangle T by 90° clockwise about the origin O.
Label the image T', and write down the coordinates of the vertices A', B', C'.

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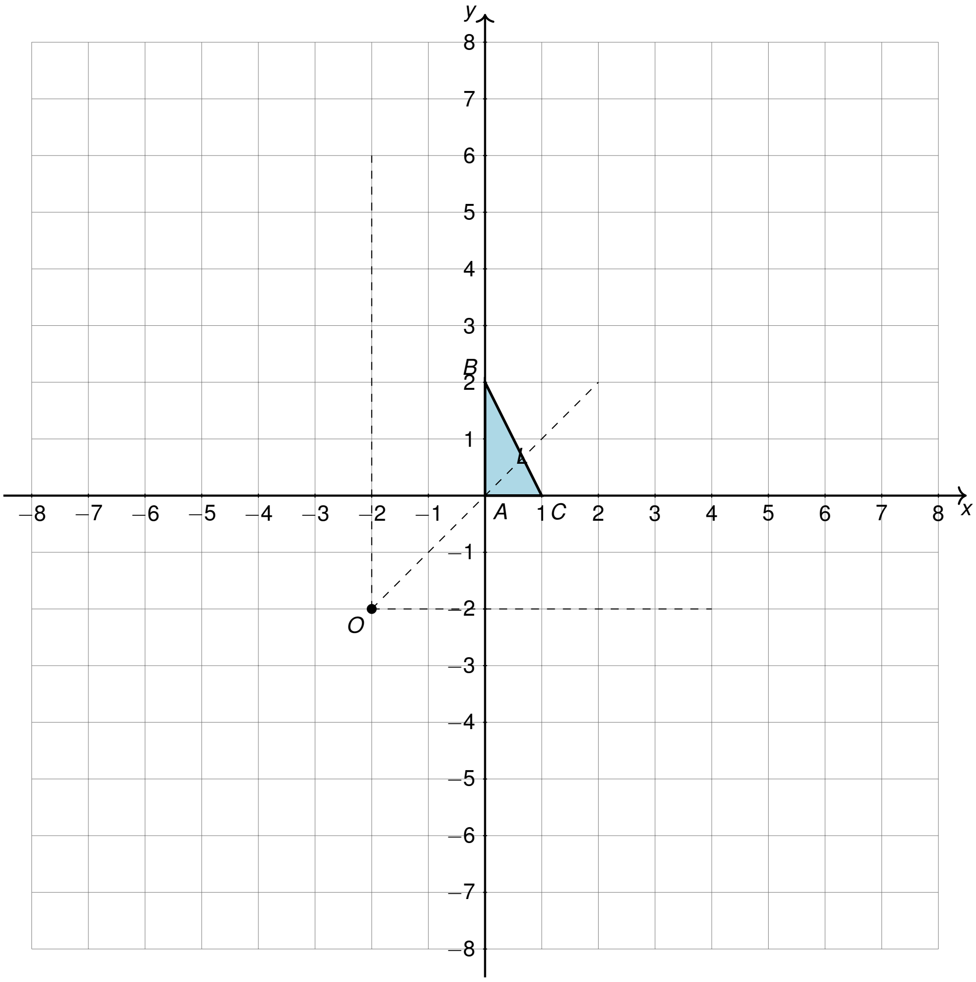
(3)

(Total for Question 4 is 3 marks)

5 A space exploration exhibit designer is enlarging a triangular logo, shape L, on a display grid to create a larger version, shape L', using centre of enlargement O.



On the grid, O is the centre of enlargement at $(-2,-2)$, and triangle L has vertices $A(0,0)$, $B(0,2)$ and $C(1,0)$.

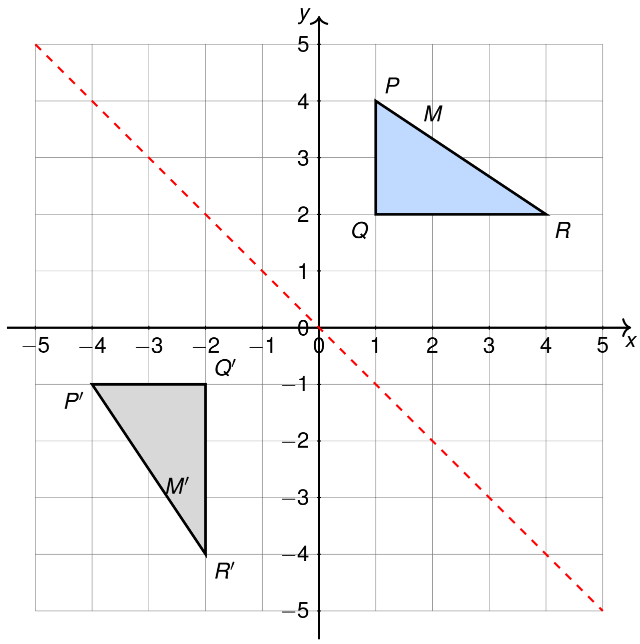
Enlarge triangle L by scale factor 2, using O as the centre of enlargement. Label the image L'.

....

(3)

(Total for Question 5 is 3 marks)

6 A space mission archivist is cataloguing star-chart overlays. Shape M represents a scanned constellation marker, and shape M' represents the same marker found on an older, mirrored chart fragment. Shape M' is the image of shape M under a single reflection.



Describe fully the single transformation that maps shape M onto shape M'.

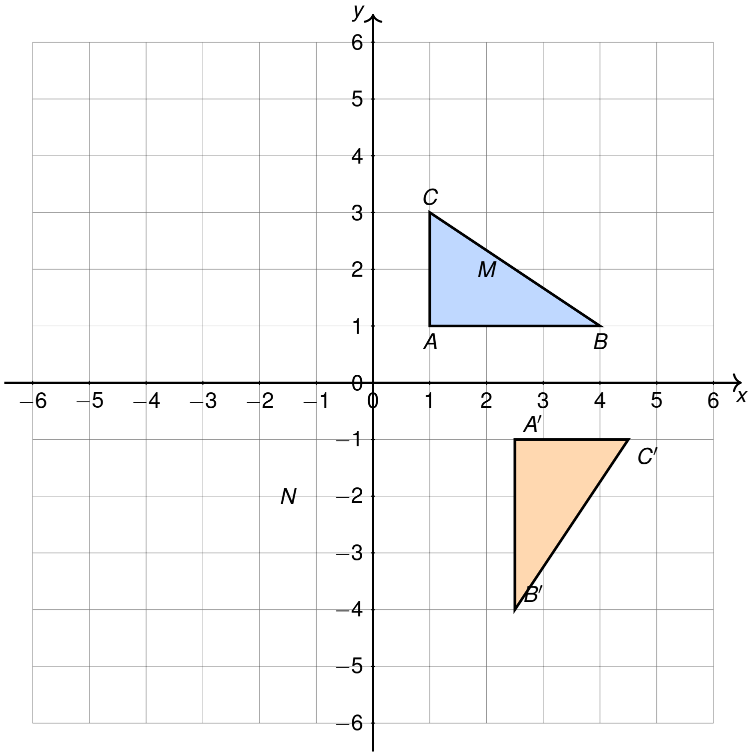
....

....

(2)

(Total for Question 6 is 2 marks)

7 Shape *M* (a mission-control observation panel, shaped as a triangle) is rotated to give shape *N*, as shown on the grid.



Describe fully the single transformation that maps shape *M* onto shape *N*.

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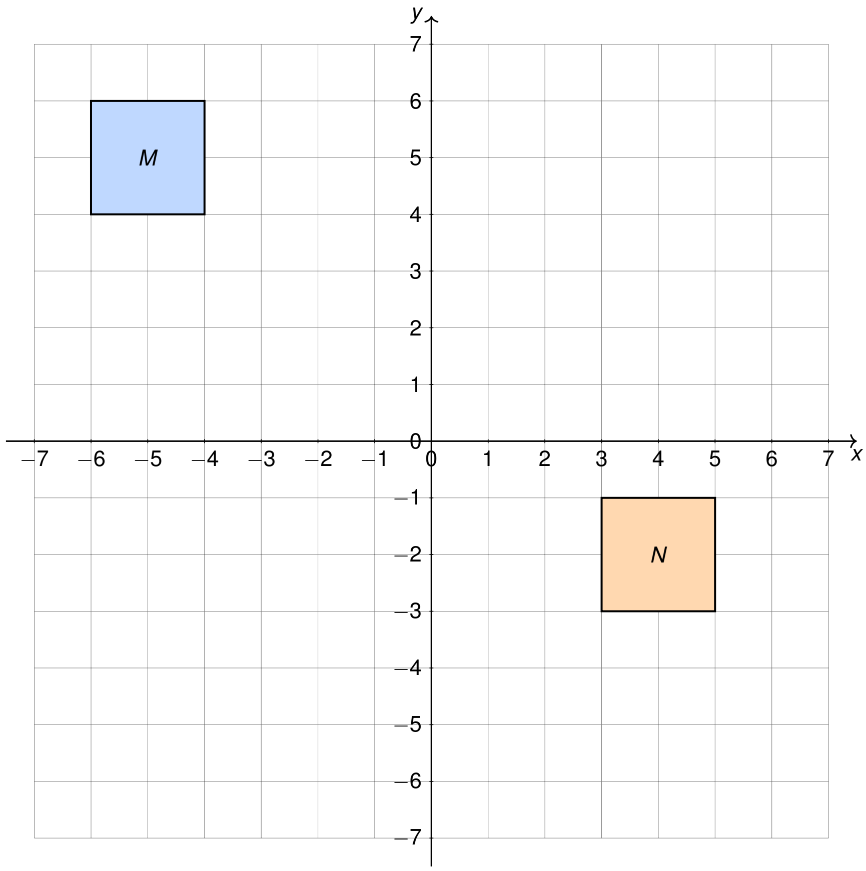
....

(3)

(Total for Question 7 is 3 marks)

8 A space agency uses a grid to model the movement of a rover exploring a rectangular survey site on the surface of Mars. Shape *M* is translated to shape *N*, as shown on the grid below.



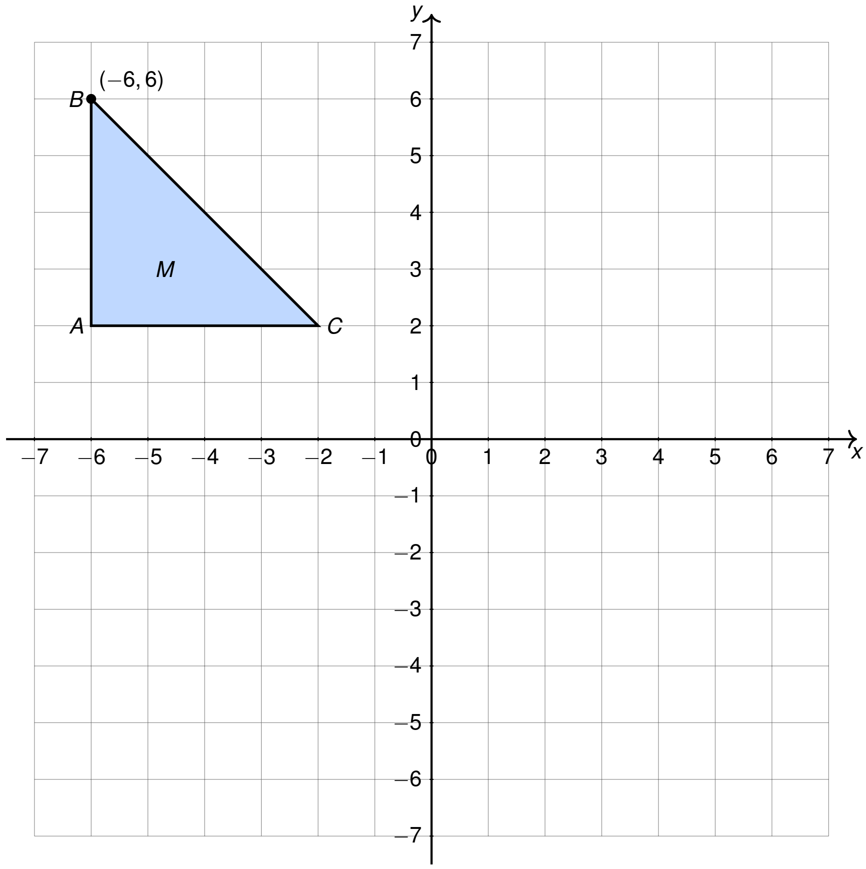
Describe fully the single transformation that maps shape *M* onto shape *N*.

....

(2)

(Total for Question 8 is 2 marks)

9 A space agency designs a scale model of a satellite dish support frame, shown as shape *M* on the grid below. The full-size frame will be built by enlarging the model with a centre of enlargement at point $(-6, 6)$ and a scale factor of $\frac{1}{2}$.

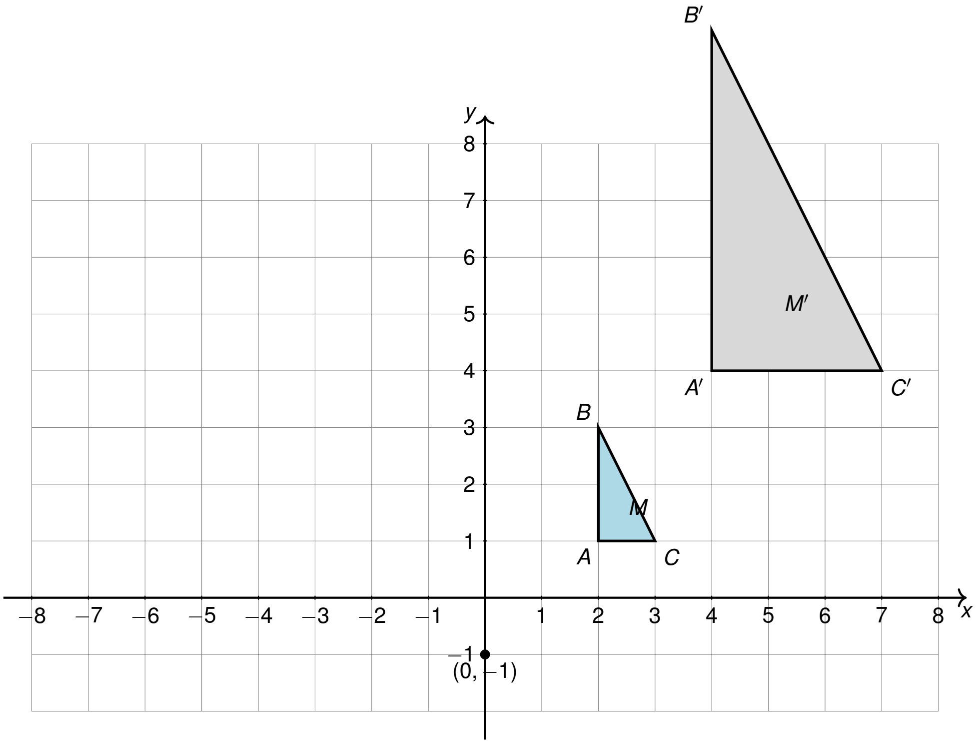


On a copy of the grid, draw the image of shape *M* after an enlargement with scale factor $\frac{1}{2}$ and centre of enlargement $(-6, 6)$.
Label the image M'.

(3)

(Total for Question 9 is 3 marks)

10 A space mission designer is scaling up a diagram of a satellite dish component for a museum exhibit. Shape M is enlarged to give shape M', as shown on the grid below.



Describe fully the single transformation that maps shape M onto shape M'.

....

....

....

(3)

(Total for Question 10 is 3 marks)

11 A space biology student is using a digital microscope to examine a rectangular sample of dried algae taken from a life-support experiment, similar to those used on the International Space Station. When photographed at $\times 1$ magnification, the rectangular outline of the sample measures 0.8 mm by 0.5 mm.

To study the cell structure more closely, the student enlarges the image so that the longer side of the rectangle now measures 6.4 mm on the screen.

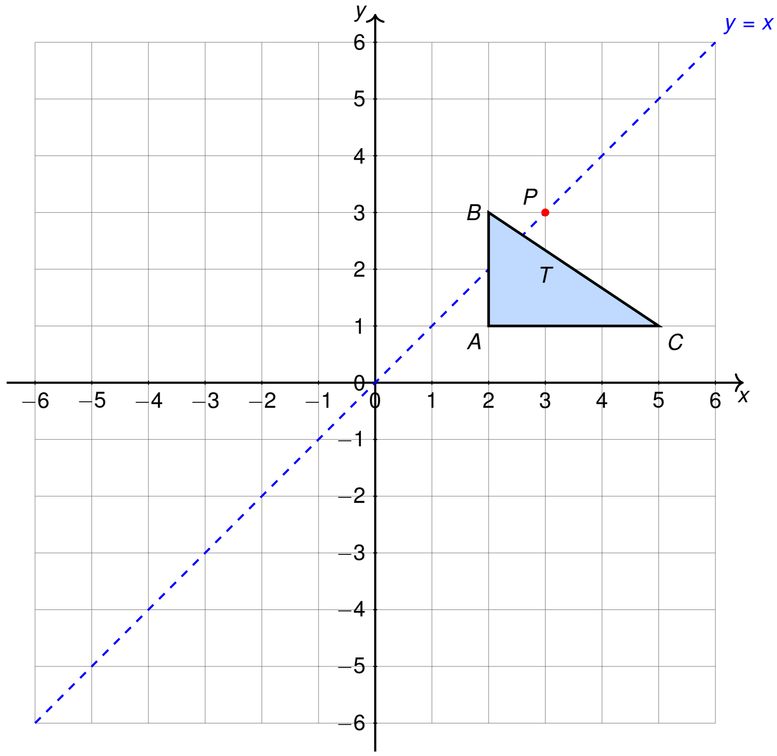
Work out the area of the enlarged rectangular image, in mm^2.

.... mm^2

(3)

(Total for Question 11 is 3 marks)

12 A historian is studying an old star-chart in which a triangular constellation marker T, with vertices $A(2, 1)$, $B(2, 3)$ and $C(5, 1)$, is reflected in the mirror line $y = x$ to produce an image marker T'.



The point P has coordinates $(3, 3)$ and lies on the mirror line $y = x$, as shown on the diagram.

Show that P is an invariant point under this reflection, i.e. show that P maps to itself.

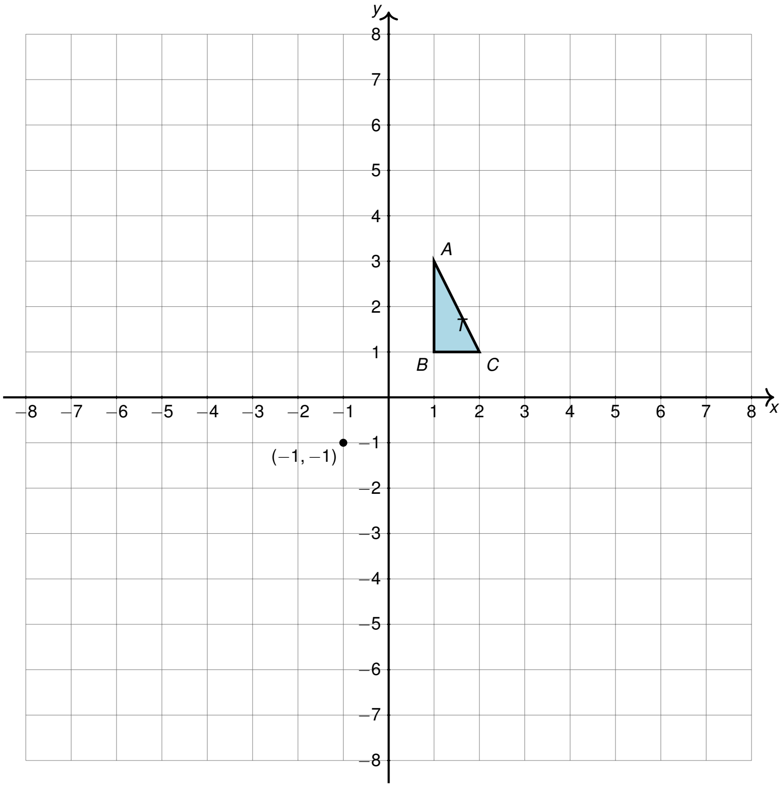
(3)

(Total for Question 12 is 3 marks)

13 A space mission designer is drawing scale diagrams of a triangular solar panel bracket, labelled T, before and after a design change that enlarges it about a fixed pivot point.

Triangle T has vertices $A(1, 3)$, $B(1, 1)$ and $C(2, 1)$.

Triangle T is enlarged with scale factor -2, centre of enlargement $(-1, -1)$, to give triangle T'.



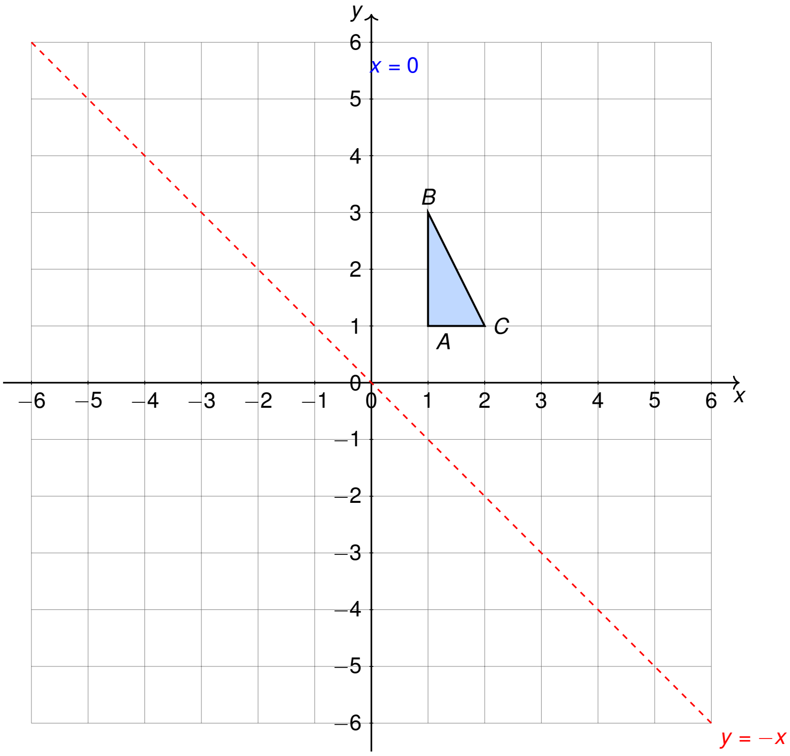
On a copy of this diagram, draw and label triangle T', the image of T after this enlargement, and state the coordinates of its vertices A', B' and C'.

....

(4)

(Total for Question 13 is 4 marks)

14 A space agency designs a decorative floor emblem for its mission-control building. The emblem starts as triangle T with vertices $A(1,1)$, $B(1,3)$, $C(2,1)$, shown on the grid below.



(a) Reflect triangle T in the line $x = 0$ (the y-axis) to give triangle T_1. Then reflect triangle T_1 in the line $y = -x$ to give triangle T_2.
On a copy of the grid (or by calculating coordinates), find the coordinates of the vertices of triangle T_2.

....

(2)

(b) Describe **fully** the single transformation that maps triangle T directly onto triangle T_2.

(2)

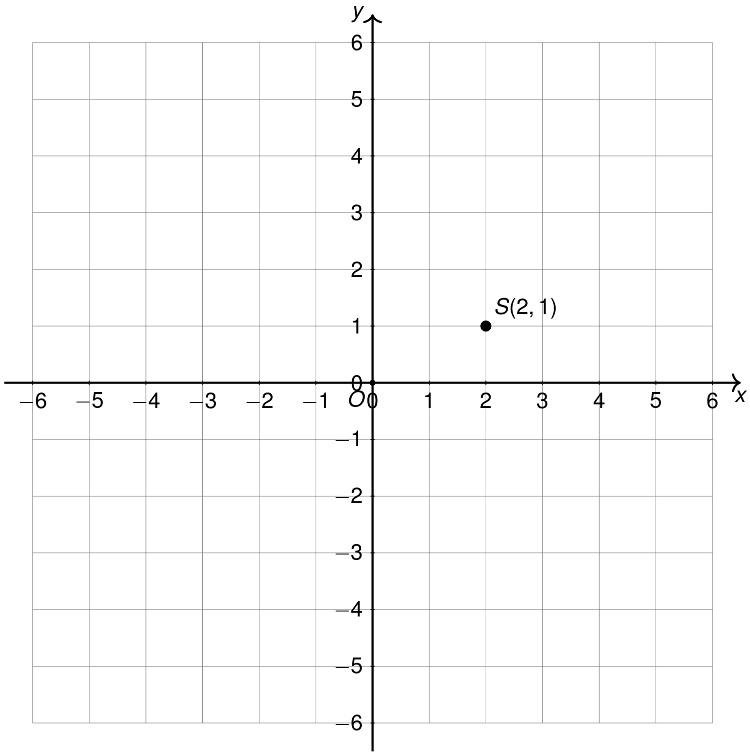
(Total for Question 14 is 4 marks)

15 Mission control is tracking a satellite as it moves through two stages of a course correction, plotted on a coordinate grid where each unit represents 100 km.

The satellite starts at point $S(2, 1)$.

Stage 1: The satellite fires its thrusters and translates by the vector $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$ to reach point S'.

Stage 2: The satellite then performs a rotation of 90° clockwise about the origin O to reach its final tracking position S''.



(a) Find the coordinates of S', the position of the satellite after Stage 1.

(........ , ...)

(1)

(b) Find the coordinates of S'', the final tracking position of the satellite after Stage 2 is applied to S'.

(........ , ...)

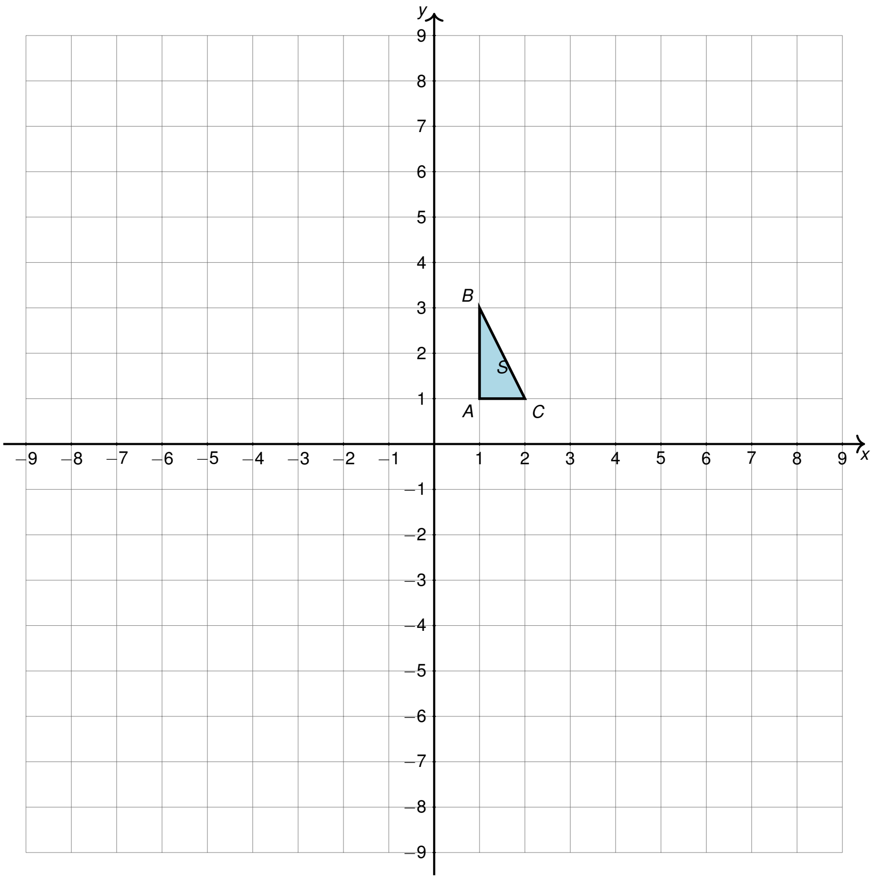
(2)

(c) A junior analyst claims that the overall journey from S to S'' could instead be described as a single rotation about the origin. Explain whether the analyst is correct, giving a reason.

(1)

(Total for Question 15 is 4 marks)

16 A space agency uses a scale model of a triangular solar panel, S, when planning the layout of panels on a satellite. Triangle S has vertices $A(1, 1)$, $B(1, 3)$ and $C(2, 1)$, as shown on the grid below.



(a) Triangle S is enlarged by scale factor 3, centre the origin $(0, 0)$, to give triangle S'.
Find the coordinates of the vertices of S'.

....

(2)

(b) Triangle S' is then translated by the vector $\begin{pmatrix} -2 \\ 5 \end{pmatrix}$ to give triangle S''.
Find the coordinates of the vertices of S''.

....

(1)

(c) The engineers need to know the final position of the corner of the panel that started at point C, after both transformations have been applied.
Write down the coordinates of this point on triangle S''.

(........ , ...)
(1)

(Total for Question 16 is 4 marks)

TOTAL FOR PAPER IS ?? **MARKS**